

Amazon Ads FP&A Executive Dashboard

Total Revenue

\$510,875,987.27

Gross Profit

\$332,100,920.27

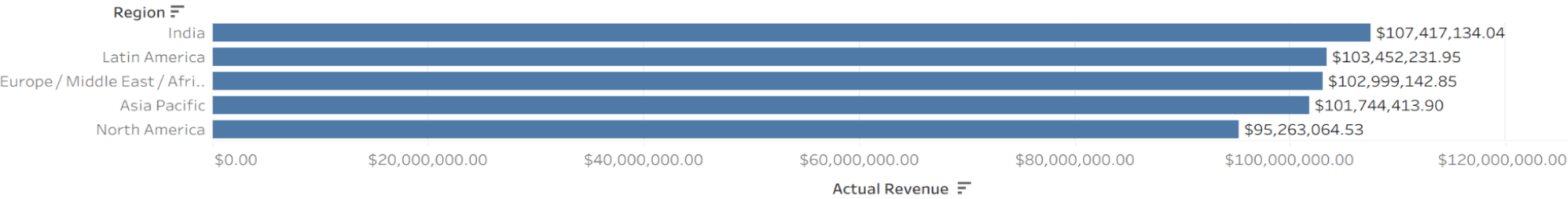
Forecast Accuracy%

101.35%

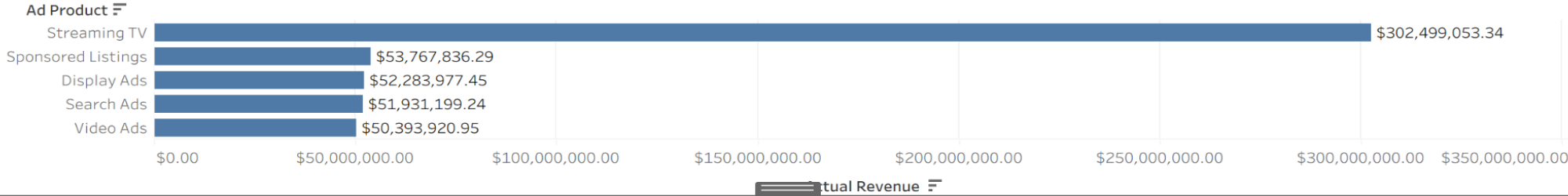
Quota Attainment %

151.82%

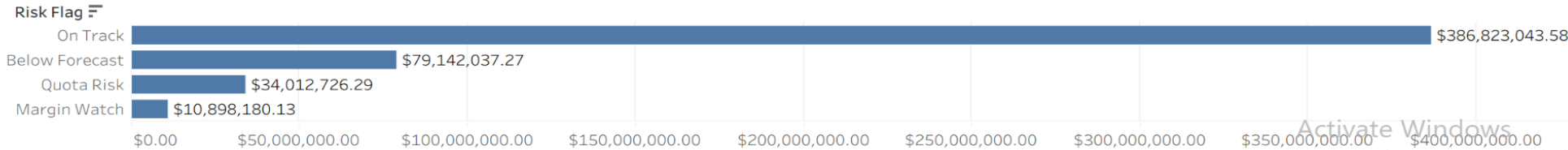
Revenue by Region



Product Performance



Risk Exposure



SQL Query:

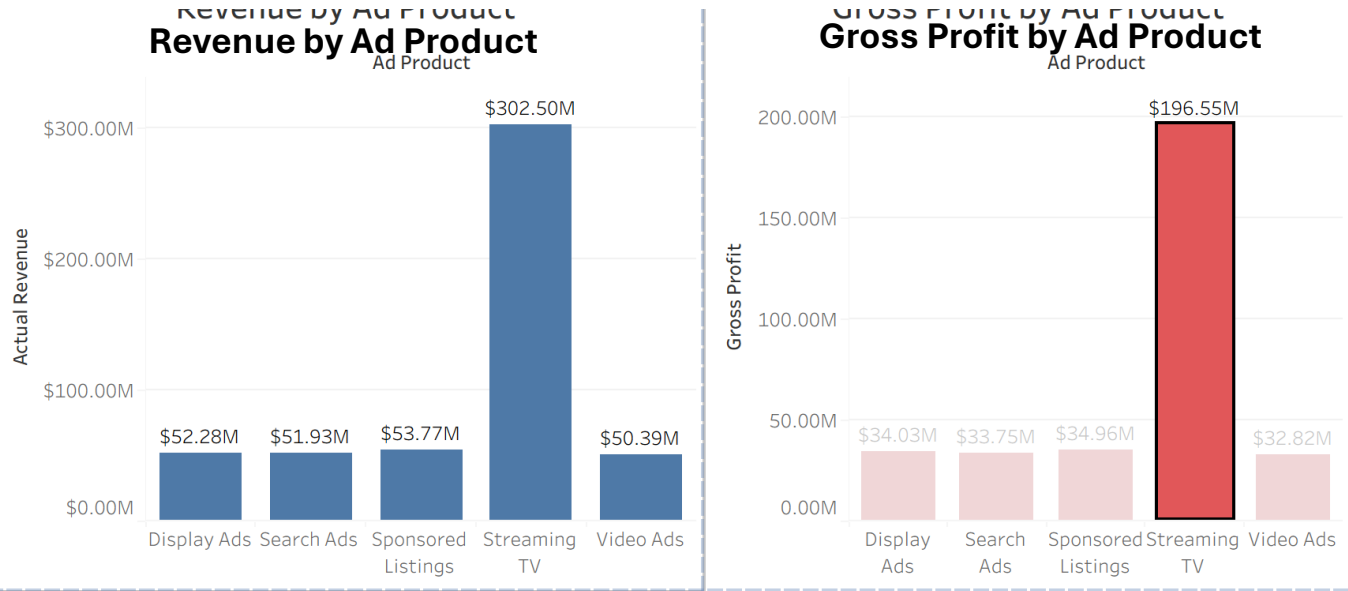
```
SELECT
  ROUND(SUM(actual_revenue),2) AS total_revenue, ROUND(SUM(gross_profit),2) AS total_gross_profit,
  ROUND(SUM(gross_profit)/NULLIF(SUM(actual_revenue),0)*100,2) AS gross_margin_pct,
  ROUND(SUM(actual_revenue)/NULLIF(SUM(forecast_revenue),0)*100,2) AS forecast_accuracy_pct,
  ROUND(SUM(actual_revenue)-SUM(forecast_revenue),2) AS revenue_variance,
  ROUND(SUM(actual_revenue)/NULLIF(SUM(quota),0)*100,2) AS quota_attainment_pct
FROM fact_ads_perf;
```

Explanation

This dashboard provides an executive summary of Amazon Ads business performance. It displays the most important KPIs including Total Revenue, Gross Profit, Forecast Accuracy, and Quota Attainment. From a single dashboard, business leaders can quickly evaluate financial performance, compare regional revenue, monitor product performance, and identify revenue risk.



Revenue by Ad Product & Gross Profit Analysis

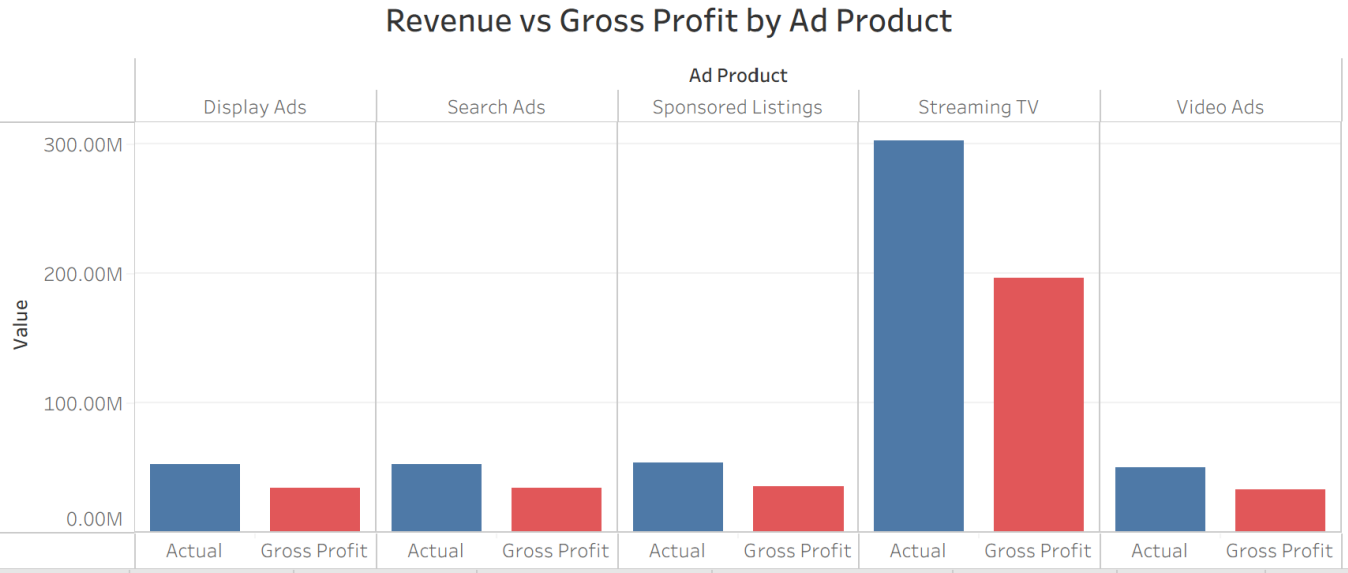


SQL Query:

```
SELECT
  ad_product,
  ROUND(SUM(actual_revenue),2) AS revenue,
  ROUND(SUM(gross_profit),2) AS gross_profit
FROM fact_ads_perf
GROUP BY ad_product
ORDER BY revenue DESC;
```

Explanation

This analysis compares revenue and gross profit across different advertising products. Streaming TV generated the highest revenue and gross profit, making it Amazon Ads' strongest performing product. This helps management understand which advertising products contribute the most to business growth.



Productivity vs Revenue by Region

Region	Avg. Productivity Score	Actual Revenue
Europe / Middle East / Afri..	122	\$102,999,142.85
India	120	\$107,417,134.04
Asia Pacific	119	\$101,744,413.90
Latin America	117	\$103,452,231.95
North America	110	\$95,263,064.53

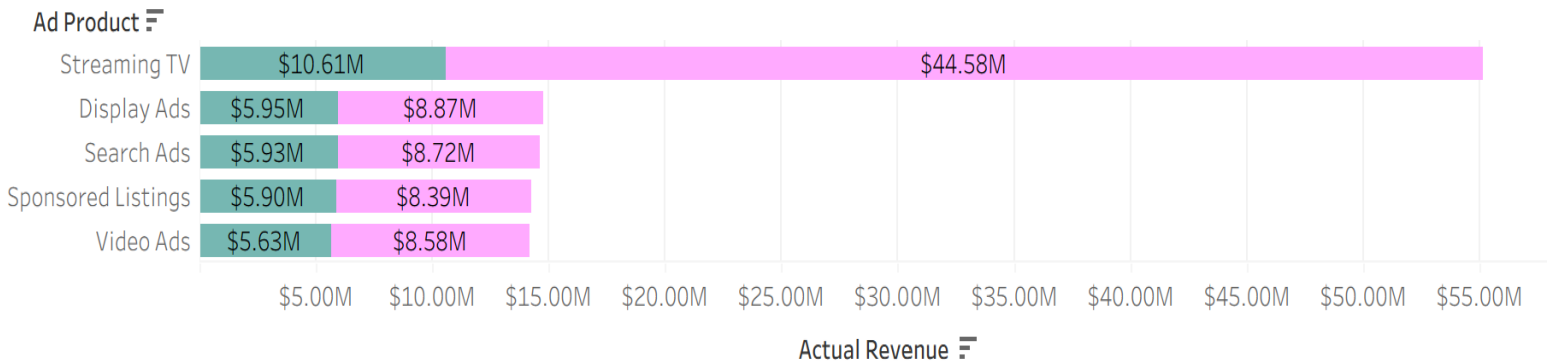
Explanation

This analysis compares average sales productivity with total revenue across regions. Although India generated the highest revenue, Europe/Middle East/Africa achieved the highest productivity score. This shows that the region generating the most revenue is not always the most efficient.

SQL Query:

```
SELECT
  region,
  ROUND(AVG(productivity_score),2) AS avg_productivity_score,
  ROUND(SUM(actual_revenue),2) AS actual_revenue
FROM fact_ads_perf
GROUP BY region
ORDER BY avg_productivity_score DESC;
```

Revenue at Risk



Risk Flag

Below Forecast

Quota Risk

SQL Query:

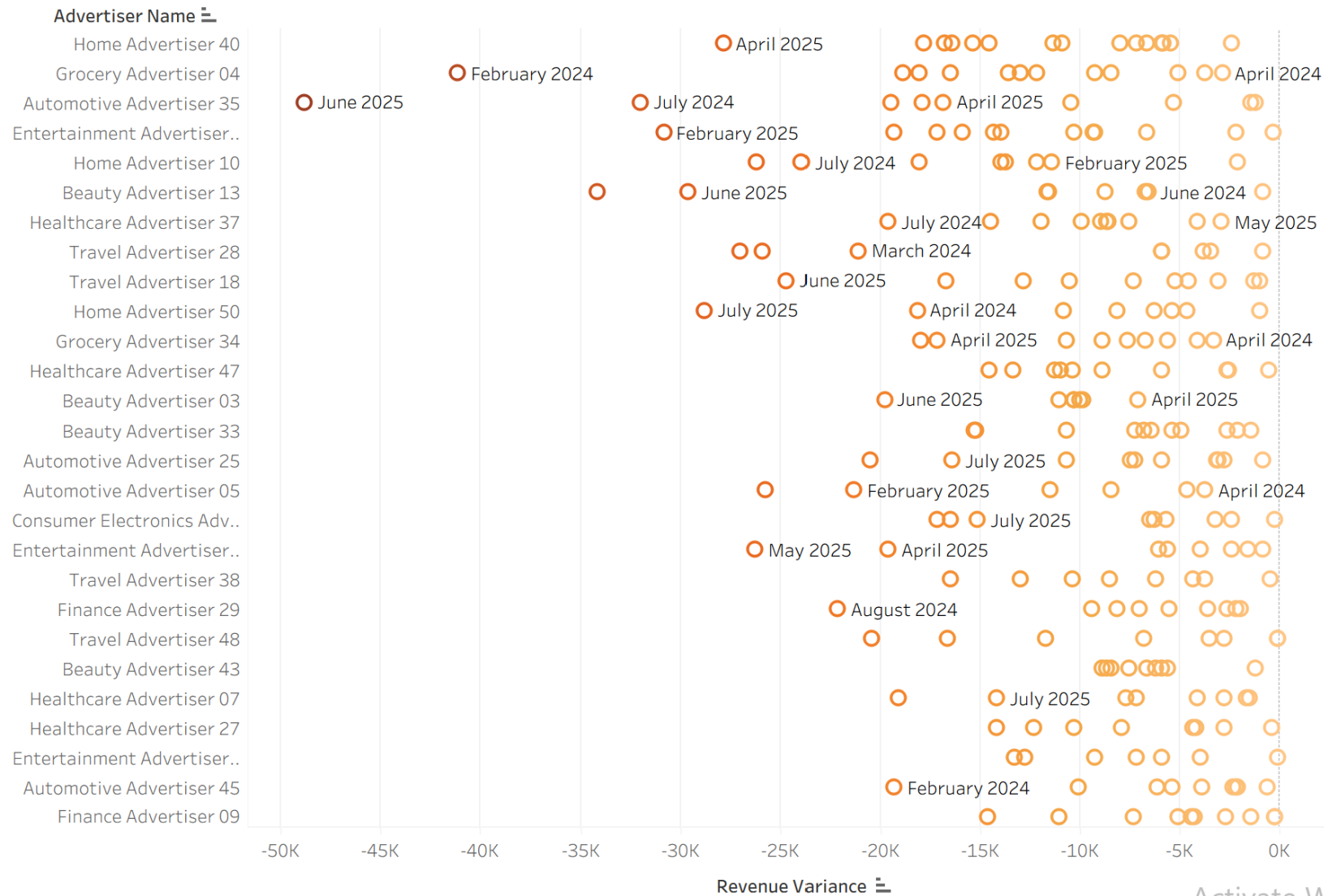
```
SELECT
  ad_product,
  ROUND(SUM(actual_revenue),2) AS revenue,
  ROUND(SUM(gross_profit),2) AS gross_profit,
  risk_flag
FROM fact_ads_perf
WHERE risk_flag IN ('Below Forecast','Quota Risk')
GROUP BY ad_product,risk_flag
ORDER BY revenue DESC;
```

Explanation

This analysis identifies revenue associated with accounts flagged as **Below Forecast** or **Quota Risk**. Streaming TV had the highest revenue exposure, indicating that while it is the company's largest revenue driver, it also requires the greatest management attention to reduce business risk.



Top Monthly Revenue Variance Shortfalls



SQL Query:

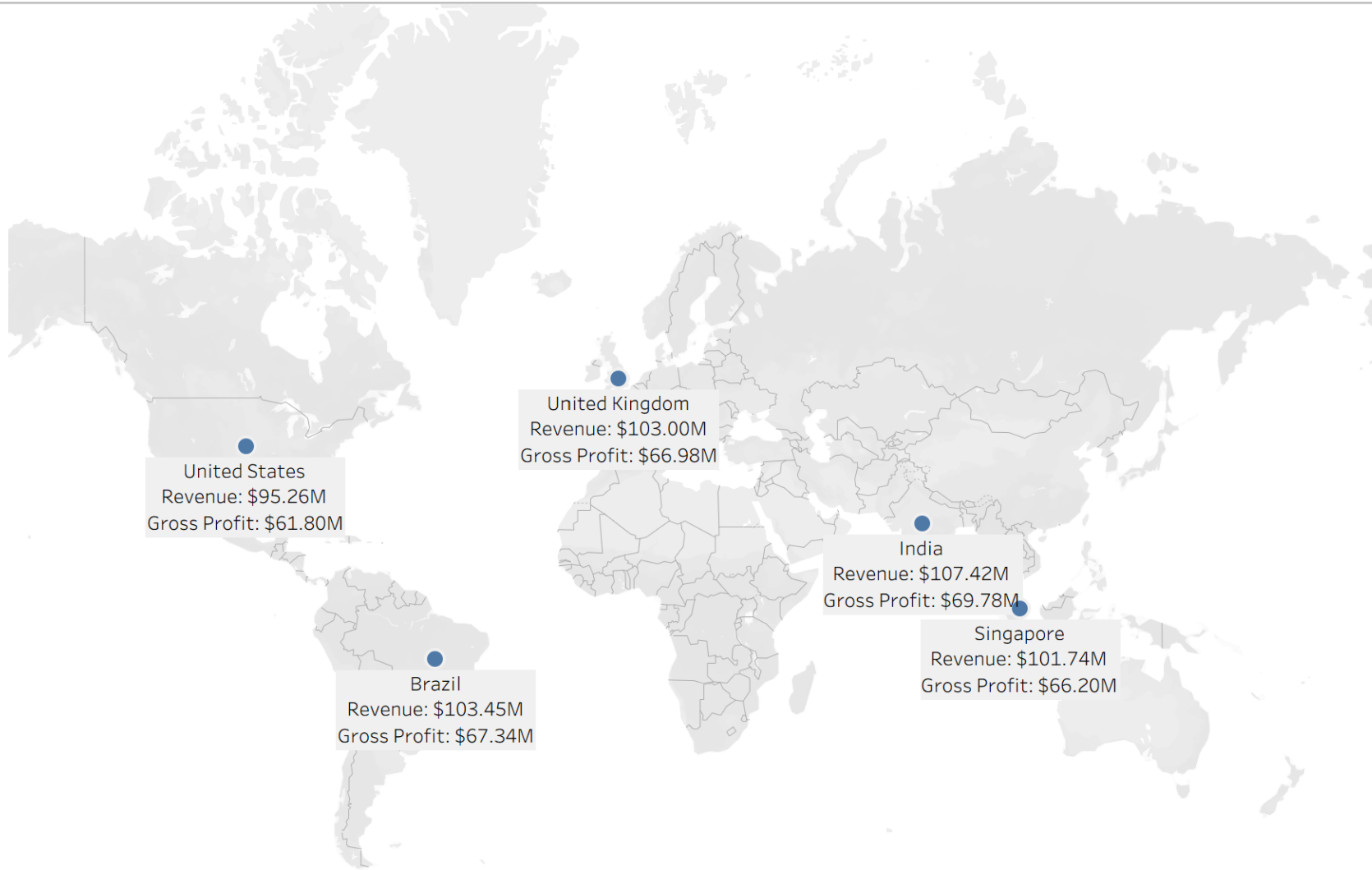
```
SELECT
    advertiser_name,
    month_date,
    revenue_variance
FROM fact_ads_perf
WHERE revenue_variance < 0
ORDER BY revenue_variance ASC;
```

Explanation

This visualization identifies advertisers that experienced the largest monthly revenue shortfalls compared to forecast. Instead of looking only at annual performance, it highlights specific months where advertisers underperformed, helping finance and sales teams investigate the causes and take corrective action.



Global Map Showing Location, Revenue & Gross Profit



SQL Query:

```
SELECT
  country,
  ROUND(SUM(actual_revenue),2)
  AS revenue,
  ROUND(SUM(gross_profit),2)
  AS gross_profit
FROM fact_ads_perf
GROUP BY country
ORDER BY revenue DESC;
```

Explanation

This map visualizes Amazon Ads revenue and gross profit across different countries. It helps identify the highest-performing global markets and provides an easy way to compare geographic business performance.



Project Summary:
Amazon Ads Financial Planning & Analysis (FP&A) Dashboard

Project Overview: The objective of this project was to simulate the role of a Financial Planning & Analysis (FP&A) Analyst supporting Amazon Ads by building an end-to-end financial reporting and business intelligence solution using **MySQL, Excel, and Tableau**.

Project Workflow

1. Dataset Preparation

Generated a realistic Amazon Ads business dataset (36,000 records) using Claude Sonnet. Reviewed and identified all business fields including revenue, forecast, gross profit, quota, productivity, products, advertisers, regions, and risk metrics.

2. Data Cleaning

Performed initial data cleaning in **Microsoft Excel** by checking column names, formatting, missing values, duplicates, and data consistency. Imported the cleaned dataset into **MySQL** and performed additional validation using SQL to ensure data quality and integrity.

3. Data Modeling

Organized the dataset into a professional analytical structure using a central **fact table** with supporting **dimension tables** for advertisers, sales representatives, and regions. Verified relationships and validated data before beginning analysis.

4. SQL Analysis

Developed business-focused SQL queries to analyze:

- Executive KPIs
- Revenue & Gross Profit
- Forecast vs Actual Performance
- Revenue Variance & Forecast Accuracy
- Sales Quota Attainment
- Productivity Analysis
- Product Performance
- Advertiser Profitability
- Revenue Risk Exposure
- Monthly, Quarterly & Yearly Business Reviews
- Advanced SQL concepts used:
- Window Functions (RANK, LAG)
- Common Table Expressions (CTEs)
- CASE Statements
- Aggregate Functions
- KPI Calculations

Outcome

This project demonstrates my ability to take a finance problem from raw data to executive reporting by cleaning data, building analytical SQL queries, developing interactive Tableau dashboards, and communicating business insights. It reflects the workflow and analytical thinking expected from a **Financial Analyst / FP&A Analyst** supporting business planning, forecasting, and strategic decision-making.

5. Tableau Dashboard Development

- Built an interactive Executive Dashboard containing:
- Executive KPI Cards
- Revenue by Region
- Product Performance Analysis
- Productivity vs Revenue
- Revenue at Risk
- Monthly Revenue Variance Shortfalls
- Global Revenue & Gross Profit Map

Skills Demonstrated

Financial Analysis: Revenue Analysis, Forecasting, Variance Analysis, Profitability Analysis, KPI Reporting

SQL: Joins, Aggregations, CTEs, Window Functions (RANK & LAG), CASE Statements

Data Visualization: Tableau Dashboards, Executive Reporting, Interactive Business Intelligence

Business Skills: FP&A Reporting, Financial Storytelling, Business Reviews (MBR/QBR/YBR), Decision Support, Data-Driven Insights

Key Business Insights

- Generated over **\$510M** in simulated advertising revenue.
- Achieved **\$332M** in gross profit.
- Forecast Accuracy exceeded **101%**, indicating revenue outperformed plan.
- **India** generated the highest regional revenue.
- **EMEA** achieved the highest productivity score.
- **Streaming TV** was the highest revenue and gross profit generating advertising product.
- Revenue risk analysis highlighted **Streaming TV** as the largest revenue exposure area requiring management attention.

